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MILITARY SPECIFICATION

THERMOMETER, SELF-INDICATING, BIMETALLIC, DIAL, 8-1/2 INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of self-indicating, dial-type thermometer, suitable for use in film-developing tanks of field radiographic outfits.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Fiberboard.

MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, Attn: Directorate of Medical Materiel, Code ATT, 2800 So. 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6685

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STANDARDS

FEDERAL

FED STD No. 66 - Steel: Chemical Composition and Hardenability.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout, suitable in all respects for the purpose intended and as herein specified.

* 3.1.1 Corrosion-resisting steel. The stem, head, clip and all exposed metal parts of the thermometer shall conform to AISI 300 series of Federal Standard No. 66.

3.2 Style and design. Thermometer shall be bimetallic, dial, self-indicating type, of strong and sturdy construction, and graduated with the Fahrenheit (F.) scale. The thermometer shall consist of a stem and head, and shall be provided with a metal clip and a protective carrying tube.

3.2.1 Stem. The stem shall be made of corrosion-resisting steel, as specified in 3.1.1, and shall be not less than 6 inches long nor more than 8 inches long.

3.2.2 Head. The head of the thermometer shall be made of corrosion-resisting steel, as specified in 3.1.1, and shall contain the dial face, 2 ± 1/4 inches in diameter, which shall be perpendicular to the stem. The dial face cover shall be of clear plastic or glass.

3.2.3 Clip. Thermometer shall have a metal clip made of corrosion-resistant steel, as specified in 3.1.1, and shall be suitable for attaching the thermometer to the side of a stainless-steel film-developing tank, holding the thermometer in an upright position, with the stem immersed in the solution.

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* 3.2.4 Carrying case. A protective carrying case shall be provided; it shall be a spiral-wound fiberboard can with a metal bottom and metal screw cap closure. As an alternate, a telescope-type closure with metal top, sealed by a strip of tape that completely encircles the can, shall be acceptable. The dimensions of the case shall be such that the thermometer with cushioning and clip shall fit snugly in place. Suitable cushioning shall be provided to prevent movement of the thermometer within the case.

3.3 Range of scale. The thermometer shall have a range from at least 25°F. to 125°F., and shall cover an arc of not less than 290°.

3.3.1 Graduation marks and numerals. The thermometer shall be graduated at intervals of 1°. Each 5° mark shall be longer than the intervening marks. Each 10° shall be marked with Arabic numerals.

* 3.4 Accuracy. When tested as specified, the thermometer shall be accurate to $\pm$ 1°F from 60° to 85°F and increasing to $\pm$ 2°F at scale extremes (see 4.4.1).

3.5 Recalibration. The design shall provide means for recalibration without disassembling.

3.6 Leakage. No liquid shall enter the head, when the thermometer is tested as specified (see 4.4.2).

3.7 Identification marking. The thermometer shall be legibly and permanently marked with the name or registered trademark of the manufacturer.

3.8 Workmanship. Workmanship shall be first class throughout. Thermometers shall be free from defects which detract from their appearance or impair their serviceability.

4. *QUALITY ASSURANCE PROVISIONS*

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

* 4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

TABLE I. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-1	2.5

* 4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. Sampling for tests.

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AQL
Thermometer	Accuracy	3.4	4.4.1	S-2	*
Thermometer	Leakage	3.6	4.4.2	S-1	*

* Acceptance number shall be zero.

4.3 Examination. The thermometers shall be examined to determine compliance with all requirements contained in this specification.

* 4.3.1 Classification of defects. Examination shall be conducted in accordance with table III, classification of defects.

TABLE III. Classification of defects.

Categories	Defects*
Major	
101	Component part missing.
102	Thermometer not free of sharp edges.
103	Thermometer not free of protrusions.
104	Dial face not perpendicular to stem.
105	Dial face not clear.
106	Dial face not free of cracks.
107	Dial face not secured.

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TABLE III. Classification of defects (continued).

Categories	Defects*
Major	
108	Graduation markings not complete, correct, or legible.
109	Stem not secured to head.
110	Stem not straight.
111	Clip cannot be secured to thermometer.
112	Cover not secured to carrying case.
Minor	
201	Thermometer not free of pits or burrs.
202	Thermometer not free of rough areas.
203	Thermometer not free of scratches.
204	Identification marking not complete, correct, legible, or permanent.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The thermometers shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be considered as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accuracy. Thermometer shall be tested against a standard thermometer at three proportionally placed points. One of these points shall be in the range of 60°F. to 70°F. The standard thermometer shall be certified by the National Bureau of Standards.

4.4.2 Leakage. The head of the thermometer shall be immersed in water at a depth of 20 inches at 68°F. for 48 hours.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) thermometer, as specified, constitutes one unit of issue.

* 5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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TABLE IV. Packaging quantities.

Unit package	Intermediate package	Exterior container
1 unit	Not required	48 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be as specified in 3.2.4.

* 5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

* 5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 3; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

* 5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with the appendix of PPP-B-636.

5.2.2 Level B.

* 5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

* 5.3.1 Interior packages. Each unit package (carrying case) shall be marked in accordance with MIL-STD-129.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use. The thermometer is intended to be suitable for use in film-developing tanks of field radiographic outfits.

* 6.2 Ordering data. Procurement documents should specify the following.

(a) Tile, number and date of this specification.
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification does not include all types, classes, grades, sizes, of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet requirements.

* 6.4 This specification covers the following item listed in the Federal Supply Catalog.

National Stock Number	Item Identification
6685-00-616-7950	THERMOMETER, SELF-INDICATING, BIMETALLIC, DIAL, 8-1/2 INCH

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 03

Review activity:
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